Information Retrieval and Web Mining

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Outlines

- Models for Information Retrieval
  - Introduction
  - Relevance feedback
  - Evaluation Metrics
  - Preprocessing for documents & web pages
  - Indexing
  - Web Search
- Link Analysis
  - Social Network Analysis
  - PageRank
  - HITS
- Latent Semantic Indexing

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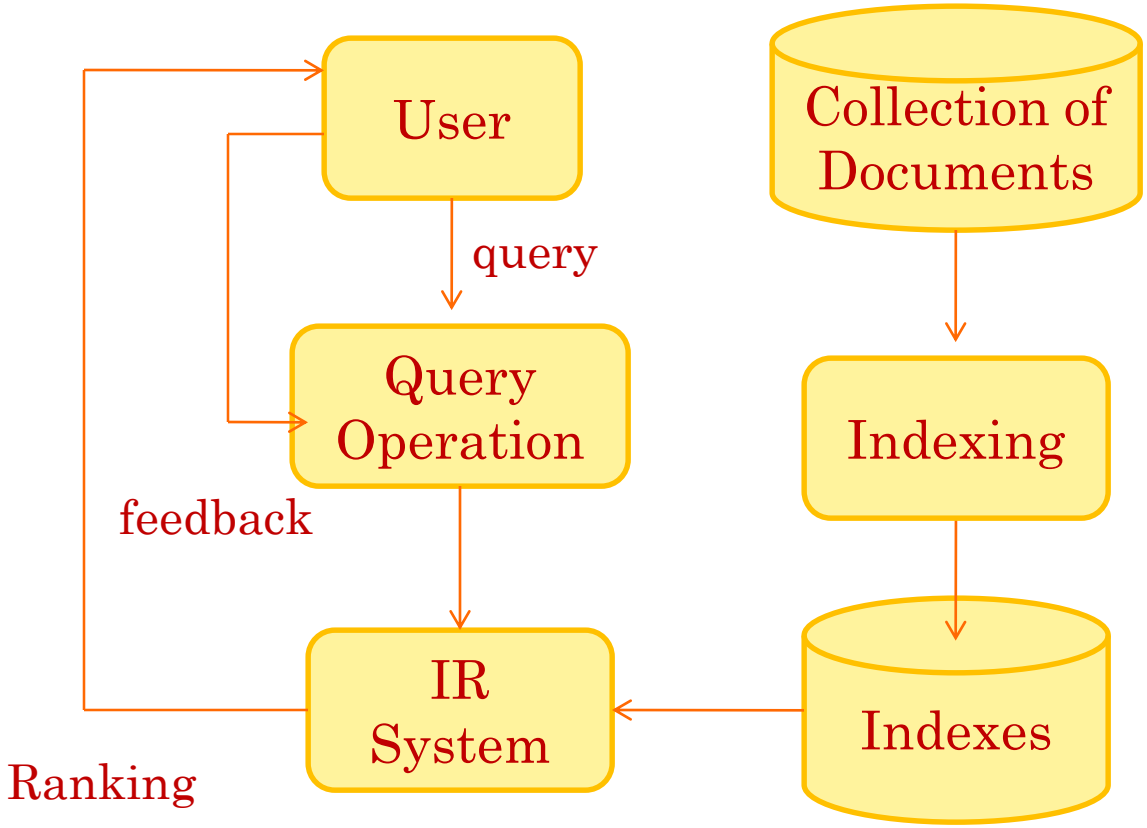
Information Retrieval

- Goal: to search data in large collection of documents
  - Text documents is unstructured
  - Documents in the collection
  - Query issued by the user
  - Keywords or terms
- Web Search
  - Application domain of IR
  - Huge amount of documents: efficiency is the critical issue
  - Web pages are semi-structured: hyperlink might contain useful information
  - Spamming needs to be overcome

Architecture of IR systems



Types of Queries

- keywords, terms
  - web mining: web + mining
- Boolean query (with AND, OR, NOT)
  - web mining OR data mining
- Phrases
  - data mining applications
- Proximity (query expansion)
  - term proximity (e.g. joyful vs. happy, synonyms)
  - term order (e.g. web mining, mining the web)
- Retrieval by examples
- Natural language question and answering (QA)

Query Operations

- Removing stop words
  - Stop words: short terms in English such as *the, that, at, in* and *a*, etc.
  - Stop words often degrade the search performance.
- User's feedback (optional)
  - Relevance feedback (e.g. clicking the items that the user feel interesting or relevant)
  - Expanding and refining the query
- Natural language processing
  - Parsing & understanding the query/contents
  - QA system

Representation of Documents

- *A bag of words* (or terms)
  - Without considering *the order of the words* appearing in the documents
  - Words might represent symbols (! ? ☺ 2012 y2k…)
- Definition
  - D: a set of documents
  - $V = \{ t_1, t_2, \ldots, t_{|V|} \}$ vocabulary set
  - |V| is the number of elements in V
  - A document vector $\boldsymbol{d_j} = (w_{1j}, w_{2j}, \ldots, w_{|V|j})$
    w_{ij}: the weight of term t_i in the j-th document.
  - Query can be represented as a vector in the same form of document vectors.

IR Models

- Boolean Model
- Vector Space Model
- Statistical Model

Boolean Model

- If term t_i occurs in the j-th document, then $w_{ij} = 1$, and 0 otherwise.
- Features
  - Without considering the occurrence count. Consider only whether it occurs or not.
  - Allow exact match only; not allow proximity match.

    (cannot provide the degree of similarity)
  - Cannot sort the documents according to relevance
  - Efficient for AND, OR, NOT operations.
  - Can be realized using the data structure *Set*

Vector Space Model

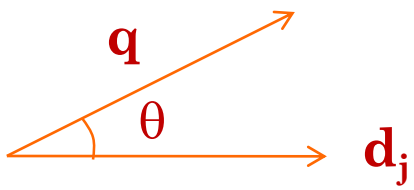
- Definition
  - f_{ij} : the occurrence count of t_i in j-th document.
  - Term frequency (TF) tf_{ij}: $tf_{ij} = \dfrac{f_{ij}}{\max\{ f_{1j}, f_{2j}, \ldots, f_{|V|j} \}}$
  - Document frequency (DF) df_i : term t_i appears in how many documents.
  - Inverse document frequency (IDF): discrimination

    $$idf_i = \log \frac{N}{df_i}$$
  - Those terms appearing in fewer documents (df small) has higher discrimination.

Weights in a Vector

- TF weight : $w_{ij} = f_{ij}$
- TF-IDF weight: $w_{ij} = tf_{ij} \times idf_i$
- Other weight (Salton and Buckley)
  - Term weight for query $\boldsymbol{q}$:
    $$w_{iq} = \left(0.5 + 0.5 tf_{iq}\right) \times idf_i$$
- Cosine similarity

$$cos(\boldsymbol{q}, \boldsymbol{d_j}) = \boldsymbol{q} \cdot \boldsymbol{d_j} / (|\boldsymbol{q}|\,|\boldsymbol{d_j}|)$$



$$\mathbf{q} \quad \mathbf{d_j}$$

$$\begin{bmatrix} \\ \\ \\ \\ \end{bmatrix} \cdot \begin{bmatrix} w_{1j} \\ w_{2j} \\ \\ \\ w_{|V|j} \end{bmatrix}$$

Measures of Relevance

- *Okapi* relevance

$$okapi(d_j, q) = \sum_{t_j \in q, d_j} \ln \frac{N - df_i + 0.5}{df_i + 0.5} \times \frac{(k_1 + 1) f_{ij}}{k_1 (1 - b + b \frac{dl_j}{avdl}) + f_{ij}} \times \frac{(k_2 + 1) f_{iq}}{k_2 + f_{iq}}$$

- Pivoted normalization weighting

$$pnw(d_j, q) = \sum_{t_j \in q, d_j} \frac{1 + \ln(1 + \ln(f_{ij}))}{(1 - s) + s \frac{dl_j}{avdl}} \times f_{iq} \times \ln \frac{N + 1}{df_i}$$

Statistical Model for IR

$$P(d_j \mid q) = \frac{P(d_j)P(q \mid d_j)}{P(q)}$$

Assume $P(d_j)$'s are equivalent for all documents.

Consider $P(q \mid d_j)$ when ranking the documents

$$P(q = q_1, q_2, \ldots, q_m \mid d_j) \cong \prod_{i=1}^{m} P(q_i \mid d_j) = \prod_{i=1}^{|V|} P(t_i \mid d_j)^{f_{iq}}$$

$$P(t_i \mid d_j) = \frac{f_{ij}}{|d_j|}, |d_j| \equiv \sum_{i=1}^{|V|} f_{ij}$$

To avoid zero probability (f_{ij}=0), smoothing techniques are usually applied. e. g. $P(t_i \mid d_j) = \dfrac{\lambda + f_{ij}}{\lambda |V| + |d_j|}$

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Relevance Feedback

- According to user's feedback for initial search results, every document can be a labeled as either *relevant* (D_r set) or *irrelevant* (D_{ir} set).
- *Rocchio* Method
  - Original query $\mathbf{q}$ can be used to generate a new expanded query $\mathbf{q_e}$ considering D_r and D_{ir}.

$$\mathbf{q_e} = \alpha\mathbf{q} + \beta\bar{\mathbf{d}}_\mathbf{r} - \gamma\bar{\mathbf{d}}_{ir}$$

$$\bar{\mathbf{d}}_\mathbf{r} \equiv \frac{1}{|D_r|}\sum_{\mathbf{d_r}\in D_r}\mathbf{d_r}, \bar{\mathbf{d}}_{ir} \equiv \frac{1}{|D_{ir}|}\sum_{\mathbf{d_{ir}}\in D_{ir}}\mathbf{d_{ir}}$$

Relevance Feedback: By Classifier

- *Rocchio* classification method

1. According to the feedback, a document may be labeled as *relevant* (D_r set) or *irrelevant* (D_{ir} set)
2. Compute the respective centroids (modified)

$$\mathbf{c_r} = \alpha\bar{\mathbf{d}}_\mathbf{r} - \beta\bar{\mathbf{d}}_{\mathbf{r'}}$$

$$\bar{\mathbf{d}}_\mathbf{r} \equiv \frac{1}{|D_r|}\sum_{\mathbf{d}\in D_r}\frac{\mathbf{d}}{|\mathbf{d}|}, \bar{\mathbf{d}}_{\mathbf{r'}} \equiv \frac{1}{|D-D_r|}\sum_{\mathbf{d}\in D-D_r}\frac{\mathbf{d}}{|\mathbf{d}|}$$

$$\mathbf{c_{ir}} = \alpha\bar{\mathbf{d}}_{ir} - \beta\bar{\mathbf{d}}_{ir'}$$

$$\bar{\mathbf{d}}_{\mathbf{ir}} \equiv \frac{1}{|D_{ir}|}\sum_{\mathbf{d}\in D_{ir}}\frac{\mathbf{d}}{|\mathbf{d}|}, \bar{\mathbf{d}}_{\mathbf{ir'}} \equiv \frac{1}{|D-D_{ir}|}\sum_{\mathbf{d}\in D-D_{ir}}\frac{\mathbf{d}}{|\mathbf{d}|}$$

3. Any document $\boldsymbol{d}$ can be judged to be *relevant* or *irrelevant* by comparing $cos(\boldsymbol{d}, \boldsymbol{c_r})$ and $\cos(\boldsymbol{d}, \boldsymbol{c_{ir}})$.

Pseudo Relevance Feedback

- Use N-best documents to extract important keywords and merge them into the original query to form a new query.

  (also named as query expansion)
- Assumption: N-best documents are relevant.
  - If N-best documents contain irrelevant documents, it is possible that the expanded query is irrelevant to the original query.
- It is called *pseudo* relevant feedback because the uses does NOT give any feedback for the search results in fact.

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Evaluation metrics for IR

- Assume
  - n-best documents for query q is

    $R_q = <d_1, d_2, \ldots\ldots, d_n>$
  - D_q : the set of all relevant documents for query q.
- If there are s_n relevant documents in top-n retrieval results for the query q, then

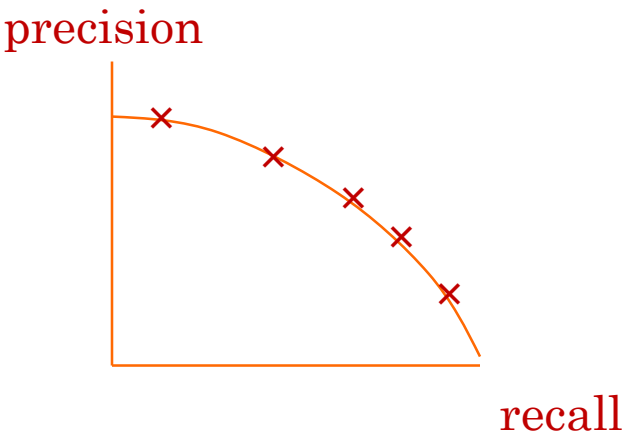
  Recall: $r(n) = s_n \ / \ |D_q|$

  Precision: $p(n) = s_n \ / \ n$
- Performance curve : precision vs. recall

Evaluation Metrics (cont'd)

$$p(r_i) = \max_{r_i \le r \le r_{10}} p(r)$$

- $p(r_i)$: the value of precision when the value of recall r is at r_i = 0%, *10%, 20%, …, 100% for i = 0, 1, 2, …, 10.*



- For multiple queries in Q, the average precision is

$$\bar{p}(r_i) = \frac{1}{|Q|} \sum_{k=1}^{|Q|} p_k(r_i)$$

Evaluation for Web Search

- Huge amount of documents: it is impossible to label *the golden truths* by inspecting all the documents
  - $|D_q|$ cannot be obtained, so *recall* cannot be computed in general
- Precision is important
  - Those results outside top-30 will not be clicked
  - precision can be obtained by inspecting the search results manually
- The *relevance* between the retrieved documents and the query is not the only concern. The *quality* and the *reputation* need also be considered.

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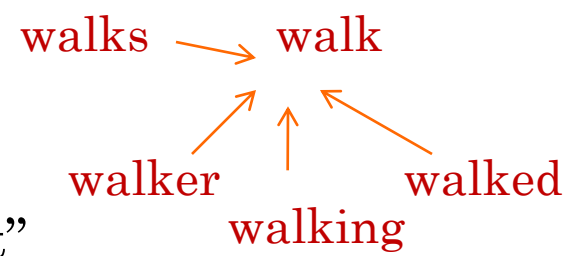
Document Preprocessing

- Remove stop words
- Convert the word into *stem* form
  - Words might be in different forms by adding prefixes or suffixes (e. g. walking, walks, walked, walker → walk)
  - Usually improve the recall, but might degrade the precision (e.g. cop, cope)
  - Can reduce the amount of indexes effectively
  - Has similar effect as clustering
- Manipulate special symbols
  - punctuations, digits, dates, times,…
  - "state-of-the-art"
    → "stateoftheart" or "state of the art"
  - Pre-processing → "preprocessing" or "pre processing"



Preprocessing for Web Pages

- Web pages are semi-structural
- Identify the areas of texts
  - title, metadata, body, …
  - area may imply different importance
- Identify *anchor text* and *HTML tags*
- Find out *main content block*
  - Banner ads, navigation bar, copyright notices might degrade the search results
  - Can use machine learning approach to identify such blocks according to visual cues (e.g. central stage)
  - HTML is a tree structure, so tree-matching can be utilized to identify main content block
  - Duplication detection is helpful
- Remove duplicated pages to reduce waste (indexing)

Duplication Detection

- *Shingles*: word n-grams in a document.
  - *Peter went to school early*

    trigram: *peter went to*, *went to school*, *to school early*
- Shingles $S_n(d)$ are important features of a document.
  - If there are too many duplicated shingles in two documents, they are likely to be the same document.
- *Jaccard coefficient* (n is the order of n-gram)

$$sim(d_1, d_2) = \frac{|S_n(d_1) \cap S_n(d_2)|}{|S_n(d_1) \cup S_n(d_2)|}$$

- More efficient to compare to documents through the shingles (could use *hash* to store the shingles).

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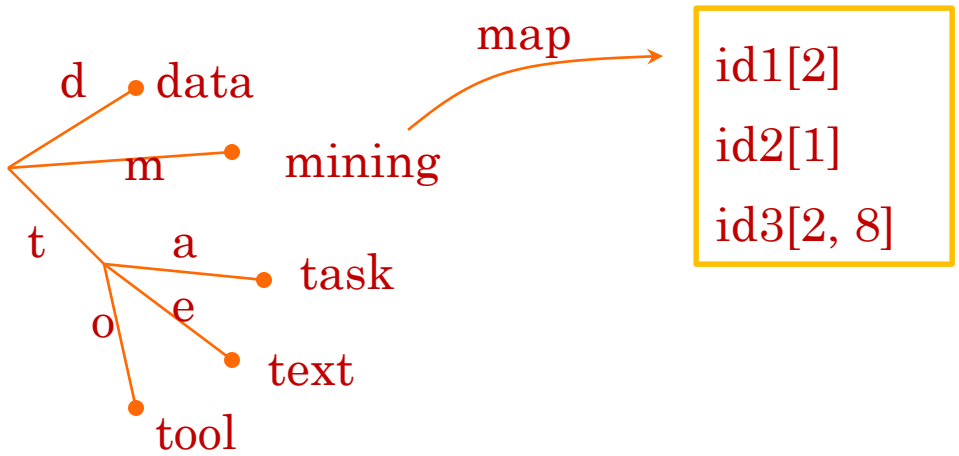
Inverted Index

- Build indexes for the keywords to refer to the original documents
  - The positions of the keywords might also be stored
- Example: three short documents
  1. Web mining is useful.
  2. Mining tools and applications.
  3. Text mining is a type of data mining.
  - mining → id1[2], id2[1], id3[2, 8]
  - web → id1[1]

Implementation of Indexes

- Suffix tree (lexical tree)



- Indexes
  - Linear structure with binary search
  - Hash table
    - JAVA: HashMap of term(String) to occurrence-set(Set)

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Issues for Web Search

- Web search is often based on vector space model and keyword matching
  - IR techniques can help to find the content-relevant pages, but the ranking of the pages is the core issue.
  - PageRank, HIT: evaluate the authorities of pages
- Multiple indexing
  - *Title* and *anchor text* contain important messages
  - Efficiency issue: the title and anchor texts can be indexed separately from and the content texts. Full-text indexes are used only if the title/anchor indexes fail to retrieve.
- Scoring (weighting) criteria
  - Type: title, anchor text, URL, body (different importance)
  - Order & position

    web mining, web … mining, mining … web

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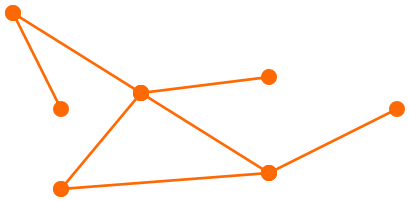
Social Network Analysis

- Degree of Centrality
  - n nodes, degree for node *i* is *d(i)*
  - Undirected graph: $C_D(i) = d(i)/(n-1)$
  - Directed graph: $C'_D(i) = d_o(i)/(n-1)$
- Closeness Centrality



$$C_C(i) = \frac{n-1}{\sum_{\substack{j=1 \\ j \neq i}}^{n} d(i, j)}$$

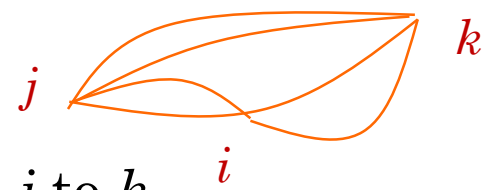
  - *d(i,j)*: shortest path from node *i* to node *j*.
  - The smaller the average distance between this node and other nodes is, the higher the closeness is.

    →this node is close to the *centroid*

Social Network Analysis

- Between-ness Centrality

$$C_B(i) = \sum_{\substack{j<k \\ j,k \neq i}} \frac{p_{jk}(i)}{p_{jk}}$$



  - p_{jk}: number of shortest paths from j to k
  - $p_{jk}(i)$: number of shortest paths from j to k through i
  - $p_{jk}(i)/p_{jk}$: the ratio of through-i paths among the shortest paths from j to k
  - Computed for C^{n-1}_2 pairs of (j,k).
  - $C_B(i)$: importance of node i as a midway node.
- After normalization

$$C_B'(i) = \frac{1}{C_2^{n-1}} \sum_{\substack{j<k \\ j,k \neq i}} \frac{p_{jk}(i)}{p_{jk}}, C_2^{n-1} = \frac{(n-1)(n-2)}{2}$$

Social Network Analysis

- Degree Prestige

$$P_D(i) = \frac{d_I(i)}{n-1}$$

More citations,
Higher prestige

The percentage of nodes that node i have influence on.

- Proximity Prestige

How importance I am within my influence domain.

  - I_i: all nodes that reaches node *i*, i.e. influence domain.
  - *d(j, i)* shortest path distance from *j* to *i*.
  - Mean shortest path $\bar{d}(i) = \frac{\sum_{j \in I_i} d(j,i)}{|I_i|}$ ($\bar{d}(i)^{-1}$ as prestige)
  - Proximity Prestige $P_P(i) = \frac{[|I_i|/(n-1)]}{\bar{d}(i)}$
  - $|I_i|/(n-1)$: ratio of nodes reaching node *i* directly or indirectly.

Social Network Analysis

- Rank Prestige
  - Prestige of a page is given by other pages
  - High prestige pages can give higher prestige to the others (e.g. recommendation letters)
- $P_R(i) = A_{1i}P_R(1)+A_{2i}P_R(2) + \ldots + A_{ni}P_R(n)$
  - $\boldsymbol{P} = \boldsymbol{A}^t\boldsymbol{P}$ ($\boldsymbol{A}$: link matrix)
  - $\boldsymbol{P}$: eigenvector corresponding to eigenvalue 1
  - link j-i: j votes i once ($A_{ji} = 0/1$)
- PageRank: with similar concept
  - $A_{ji} = 1/d_O(i)$: prestige need be distributed
  - Total amount of credit for every node is 1.0.
  - The more persons you recommend, the lower the credits everyone gets.

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PageRank Algorithm

$$\mathrm{P}(i) = \sum_{(i,j)\in E} \frac{P(j)}{d_O(j)}$$

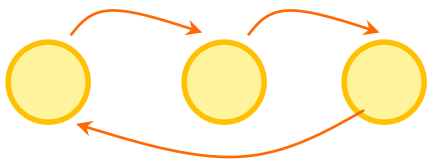
$$\mathbf{P} = (P(1), P(2), \ldots, P(n))^t$$

$$\mathbf{A}_{ij} = \begin{cases} \dfrac{1}{d_O(i)} & if\ (i,j) \in E \\ 0 & otherwise \end{cases}$$

$$\mathbf{P} = \mathbf{A}^{\mathbf{t}}\mathbf{P} \quad \mathbf{P}: principal\ eigenvector$$

$$Markov\ chain: \mathbf{P_k} = \mathbf{A}^{\mathbf{t}}\mathbf{P_{k-1}}$$

$$stationary\ probabilities\ \boldsymbol{\pi} = \lim_{k\to\infty} \mathbf{P_k}$$



periodic

- A has *stationary* probability distribution if
  - A is a stochastic matrix : NO dangling (IN only) nodes
  - A is strongly connected (irreducible)
  - A is aperiodic

PageRank Algorithm

- Add small probability (1/n)(1-d)

$$P = \left((1-d)\frac{E}{n} + dA^t \right) P$$

$$E = \begin{pmatrix} 1 & 1 & \dots & 1 \\ 1 & 1 & \dots & 1 \\ . & . & \dots & . \\ 1 & 1 & \dots & 1 \end{pmatrix}$$

- Power iteration method to approach **P**

$\mathbf{P_0} \leftarrow \mathbf{e}/n$, $k \leftarrow 1$
repeat
    $\mathbf{P_k} \leftarrow (1-d)\mathbf{e} + d\mathbf{A}^t\mathbf{P_{k-1}}$;
    $k \leftarrow k+1$
until $|\mathbf{P_k} - \mathbf{P_{k-1}}| < \varepsilon$
return $\mathbf{P_k}$

Pros and Cons of PageRank

- Benefits
  - Ability to fight *spam*
  - Global measure and query independent
  - Can be computed and saved off-line
  - Efficient at query time
- Defects
  - Cannot distinguish the authorities in *general domain* or *specific domains*
  - Static ranking: cannot provide timely information and usually reply the old pages.
  - Improvement: TimedPageRank: $d=f(t)$

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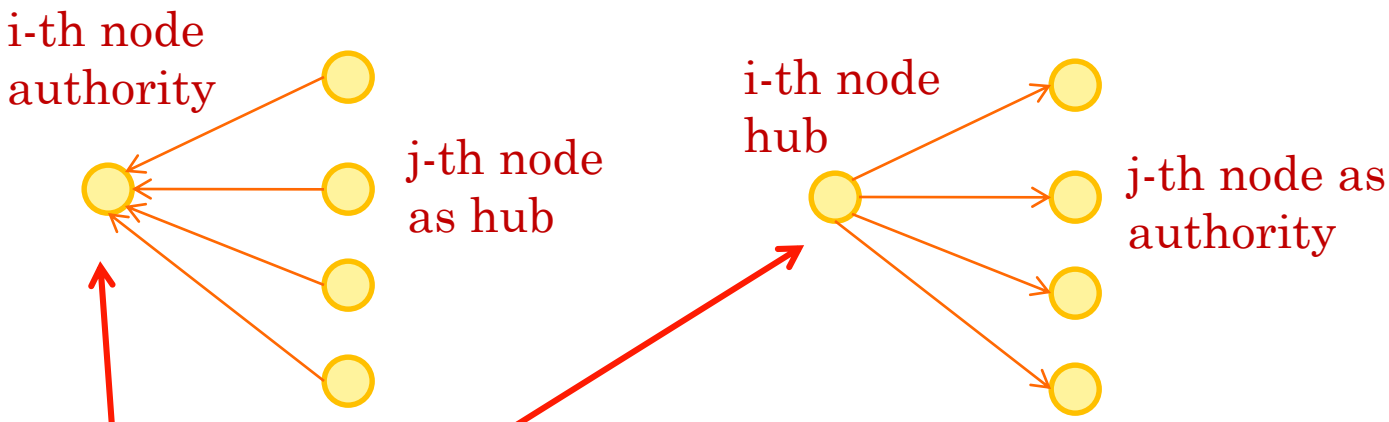
HITS Algorithm



$$a(i) = \sum_{(j,i)\in E} h(j)$$

$$h(i) = \sum_{(i,j)\in E} a(j)$$

- Valuable hubs/authorities enhance one another
- $L_{ij}=1$ *if and only if* (i,j) in E

HITS Algorithm (Cont'd)

$$a(i) = \sum_{(j,i)\in E} h(j)$$

$$h(i) = \sum_{(i,j)\in E} a(j)$$

- $L_{ij}=1$ *if and only if* (i,j) in E

$$\mathbf{a} = \mathbf{L}^{\mathbf{t}}\mathbf{h}$$

$$\mathbf{h} = \mathbf{L}\mathbf{a}$$

$$\mathbf{a_k} = \mathbf{L}^{\mathbf{t}}\mathbf{L}\mathbf{a_{k-1}} (\text{co-citation } \mathbf{C} = \mathbf{L}^{\mathbf{t}}\mathbf{L})$$

$$\mathbf{h_k} = \mathbf{L}\mathbf{L}^{\mathbf{t}}\mathbf{h_{k-1}} (\textit{Bibliographic coupling } \mathbf{B} = \mathbf{L}\mathbf{L}^{\mathbf{t}})$$

$$\mathbf{a_0} = \mathbf{h_0} = (1,1,\ldots,1)^t.$$

HITS Algorithm (Cont'd)

- Steps
  1. Search t pages(e.g. 200) as root set, W.
  2. Expand W into a relevant set according to the in-links and out-links of the pages in W. This set is called base set, S.
  3. Compute authority scores (**a**)和hub scores(**h**) for S.
  4. Select authorities/hubs to reply to the user.
- HITS-Iterate

$\mathbf{a_0}=\mathbf{h_0}=(1,1, \ldots, 1)^t$, k=1

repeat

$\mathbf{a_k} = \mathbf{L}^t\mathbf{L}\mathbf{a_{k-1}}$, $\mathbf{h_k}=\mathbf{L}\mathbf{L}^t\mathbf{h_{k-1}}$

$\mathbf{a_k} = \mathbf{a_k}/|\mathbf{a_k}|$, $\mathbf{h_k} = \mathbf{h_k}/|\mathbf{h_k}|$

k=k+1

until $|\mathbf{a_k}-\mathbf{a_{k-1}}|<\varepsilon$ and $|\mathbf{h_k}-\mathbf{h_{k-1}}| < \varepsilon$

return $\mathbf{a_k}$ and $\mathbf{h_k}$

Pros and Cons of HITS

- Benefits
  - Can rank the pages according to the query topic
  - authorities/hubs
- Defects
  - Cannot avoid spams because web designer can increase the hub scores by adding out-links in the pages.
  - *Topic drift* might occur when the root set is not good
  - Computation complexity is high
- Improvement
  - Introduction of random jump(~d in PageRank)
  - Use the concept of prestige distribution in PageRank
  - Compare the expanded pages and original query topic (for handling topic drifting)
  - Consider only anchor text or use different weights

Community Discovery

- Find out highly relevant community according to some seed pages
- Steps for growing the set

  Select *seed set*

  Repeat

  1. Find the pages by in-links/out-links of the pages in seed set till specific depths
  2. Separate relevant pages and irrelevant pages with respect to seed set based on maximum-flow min-cut
  3. Add the pages relevant to the seed pages (i.e. highest ranked non-seed pages) into the seed set.

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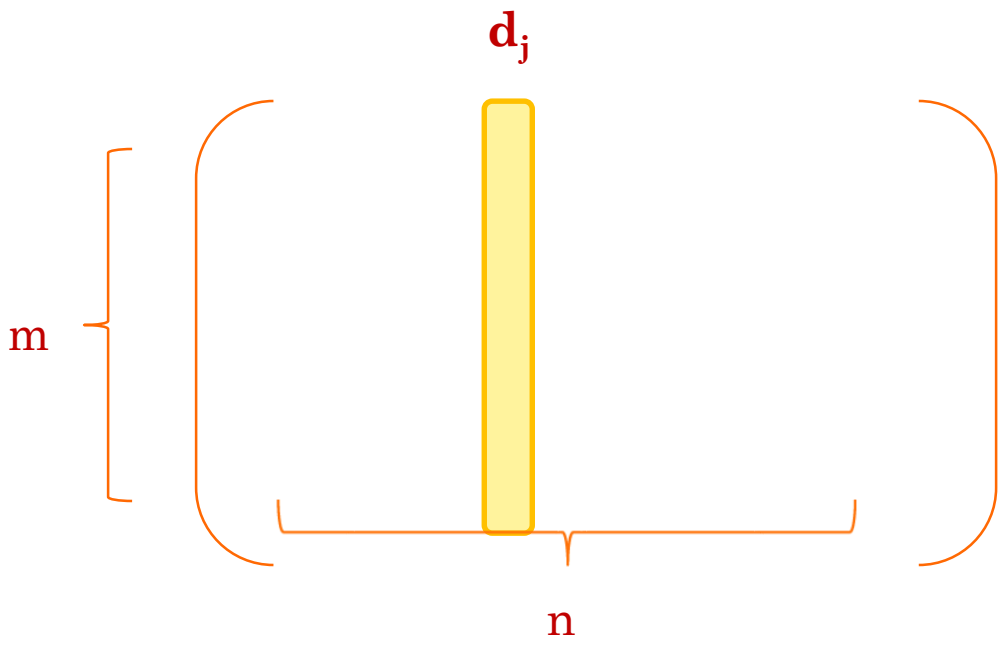
- Link Analysis
  - Social Network Analysis
  - PageRank
  - HITS
- Latent Semantic Indexing

Latent Semantic Indexing (LSI)

- Assume there are n documents containing m words. All vectors of documents can be represented as an m × n matrix **A**, called the *term-document matrix*.
- *j*-th column vector of matrix **A** is document $\boldsymbol{d}_j$.



Singular Value Decomposition

- $\boldsymbol{A}_{m\times n} = \boldsymbol{U}_{m\times r}\, \boldsymbol{\Sigma}_{r\times r}\, \boldsymbol{V}^{t}_{r\times n}$
- $\boldsymbol{U}$: right singular vectors
- $\boldsymbol{V}$: left singular vectors
- $\boldsymbol{\Sigma} = diag(\sigma_1, \sigma_2, \ldots, \sigma_r)$ where $\sigma_i \geqq 0$

  σ_i : singular values (square root of eigen-values)
- Select the largest k among r singular values (corresponding to the highest variances)
- $\boldsymbol{A'} = \boldsymbol{U}_k\, \boldsymbol{\Sigma}_k\, \boldsymbol{V}_k^{t}$ (reduce to k-dimensional vector)
- Document vectors: in $\boldsymbol{V}_k^{t}$
- Term vectors: $\boldsymbol{U}_k$
- Query vector: $\boldsymbol{q}_k^{t} = \boldsymbol{q}^{t}\, \boldsymbol{U}^{t}_{k} \boldsymbol{\Sigma}_k^{-1}$

Discussions on LSI

- Convert document/term into eigen-space in which all dimensions are statistically independent.
  - In LSI, dimensions with largest variances (σ_i)will be kept while the rest dimensions are eliminated so as to reduce the computational complexity.
  - In eigen space, terms and documents are represented in the same form. Therefore, the similarity between documents or that between terms can be computed.
  - In original document space, the cosine similarity between, say, tour and travel is 0. In eigen-space, their similarity might not be 0.
  - Search , clustering, classification, and GMM training can be performed in eigne-space.
- Higher efficiency can be achieved for LSI, but the complexity of SVD is high ($O(m^2n)$) when the number of documents is large.

Applicability of LSI

- It is possible to defined similarity between concepts (not numerical data)
  - Terms: similarity(tour ~ travel) ? similarity(tour, visa)?
  - Computed not by *word senses*, but by the examples of *usage contexts* (documents) of the words
- Generalization
  - Matrix A is NOT limited to represent term-document relationships. It can be generalized for *arbitrary M-to-M mappings* (tabular attribute-value data)
  - SVD can also be performed, and so do search, clustering, classification, statistical models (e.g. GMM) etc.
  - topic/author → similarity between authors

    user/product → clustering products

    album/style → distance between albums

Reference Book

- Web Data Mining, Springer
  Bing Liu